



Faculty of Computer Applications

**BCA Sem-1
(AY-2026-27)**

Lab Book

**Sub Code : 05BC3103
(Programming Practices : 1)**

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5.	Calculate the Square and Cube of a number.
6.	Interchange the value of two numbers.
7.	Find largest out of three numbers.
8.	Find out Fibonacci series up to given n values.
9.	Find out Number is Odd or Even.
10.	Find out sum of digits of a given number.

(1) Print "Hello World".

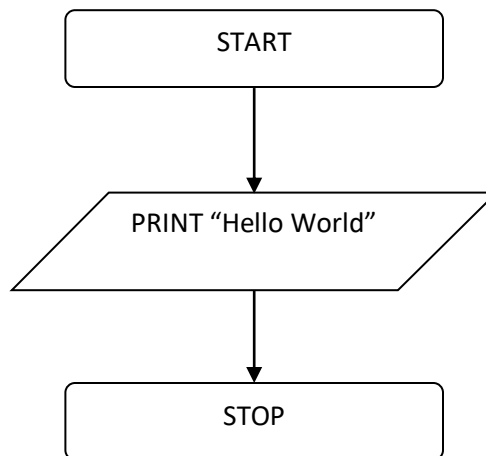
Algorithm :

Step 1 : START

Step 2 : PRINT "Hello World"

Step 3 : STOP

Flowchart :

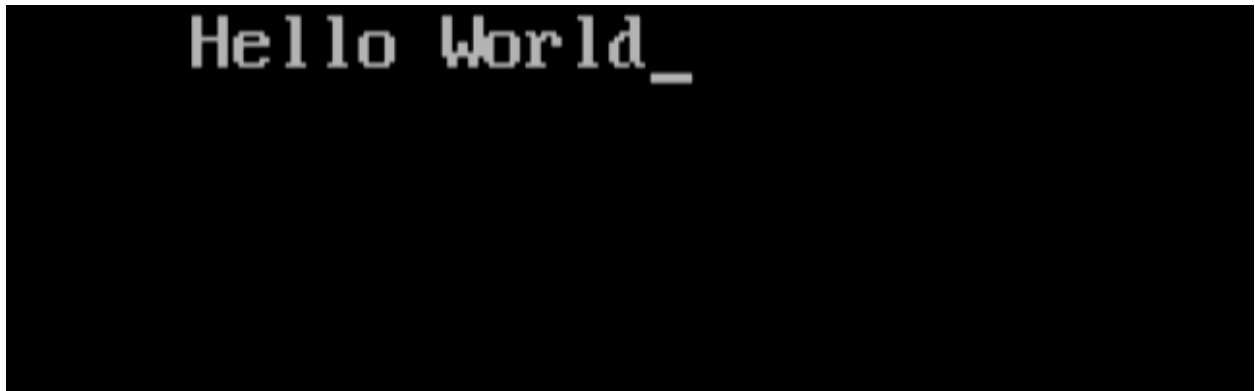


Code :

```
#include<stdio.h>
#include<conio.h>

void main(){
    clrscr();
    printf("Hello World");
    getch();
}
```

Output :

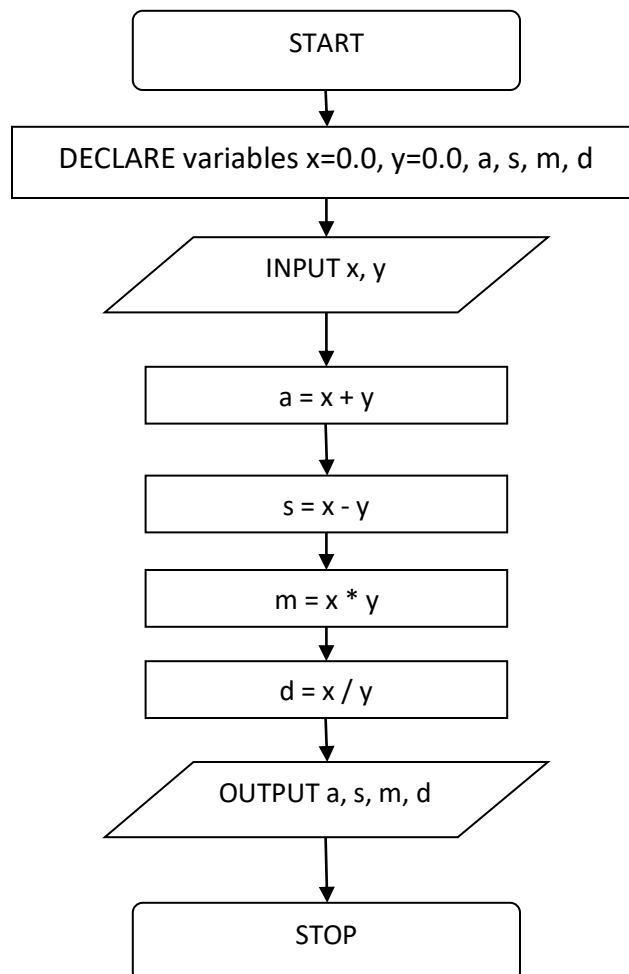
A screenshot of a terminal window with a black background. The text "Hello World_" is displayed in a light gray, monospaced font at the top left of the window. The underscore indicates that the program is waiting for a character input.

(2) 2 Values from user and Perform All Arithmetic Operations.

Algorithm :

Step 1 : START
Step 2 : DECLARE variables $x=0.0$, $y=0.0$, a , s , m , d
Step 3 : INPUT x, y
Step 4 : $a = x + y$
Step 5 : $s = x - y$
Step 6 : $m = x * y$
Step 7 : $d = x / y$
Step 8 : OUTPUT a, s, m, d
Step 9 : STOP

Flowchart :



Code :

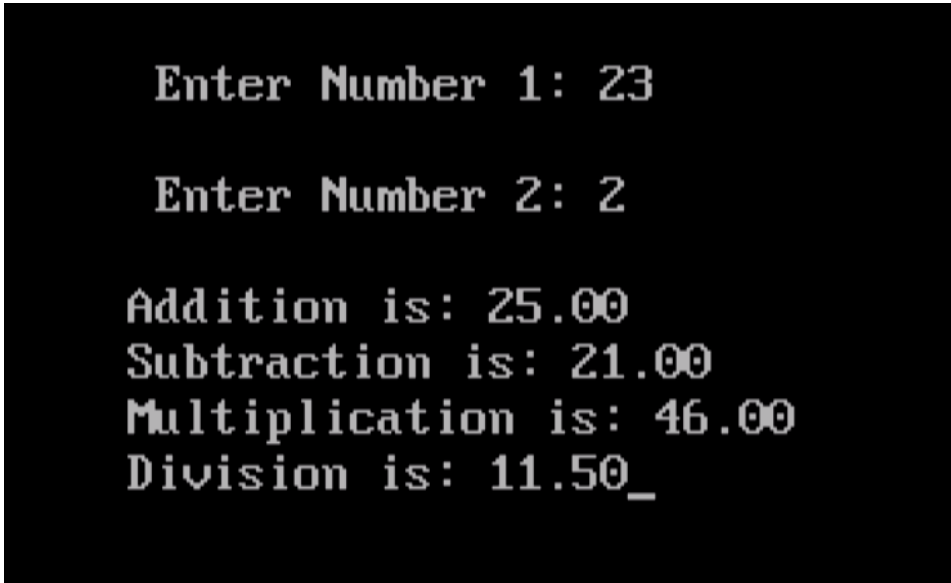
```
#include<stdio.h>
#include<conio.h>

void main()
{
    float x,y;
    clrscr();
    printf("\n Enter Number 1: ");
    scanf("%f",&x);
    printf("\n Enter Number 2: ");
    scanf("%f",&y);

    printf("Addition Is: %.2f \n",x+y);
    printf("Subtraction Is: %.2f \n",x-y);
    printf("Multiplication Is: %.2f \n",x*y);
    printf("Division Is: %.2f \n",x/y);

    getch();
}
```

Output :

A screenshot of a terminal window with a black background and white text. It shows the output of a C program. The text is: "Enter Number 1: 23", "Enter Number 2: 2", "Addition is: 25.00", "Subtraction is: 21.00", "Multiplication is: 46.00", and "Division is: 11.50_".

```
Enter Number 1: 23
```

```
Enter Number 2: 2
```

```
Addition is: 25.00
```

```
Subtraction is: 21.00
```

```
Multiplication is: 46.00
```

```
Division is: 11.50_
```

(4) Input 3 values from user and find out its Average.

Algorithm :

Step 1 : START

Step 2 : DECLARE variables x, y, z, avg

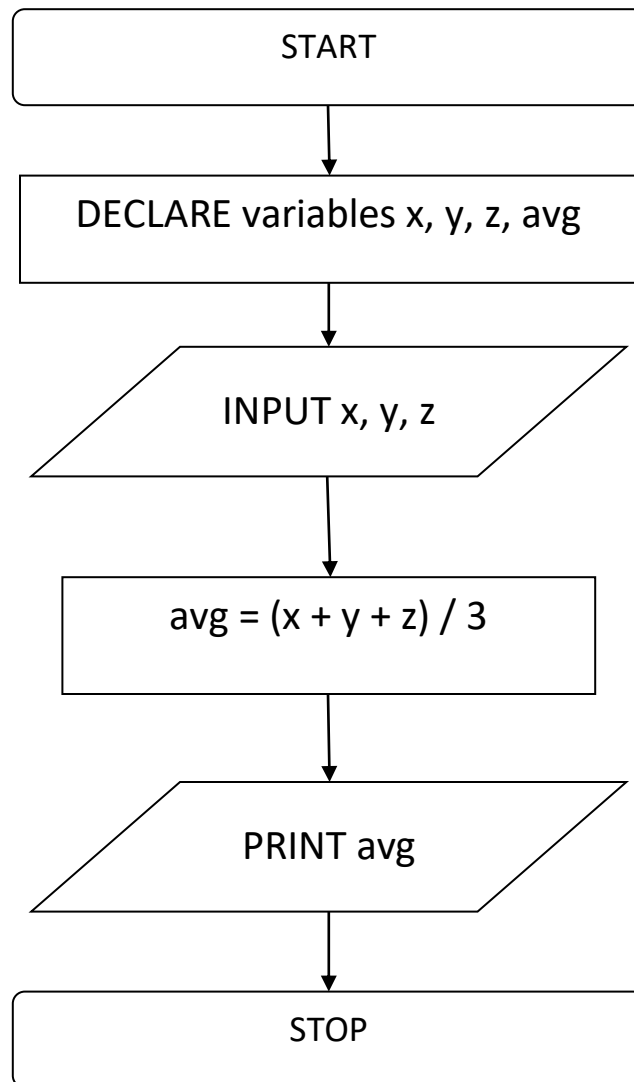
Step 3 : INPUT x, y, z

Step 4 : $\text{avg} = (x + y + z) / 3$

Step 5 : PRINT avg

Step 6 : STOP

Flowchart :

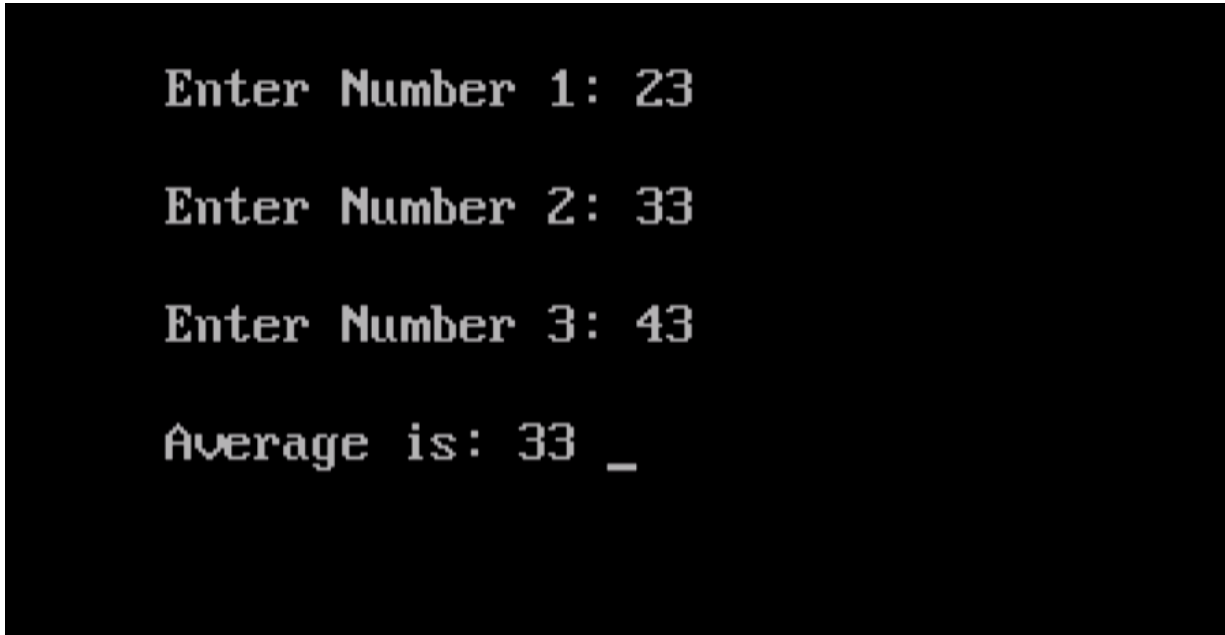


Code :

```
#include<stdio.h>
#include<conio.h>

void main()
{
    int x,y,z;
    clrscr();
    printf("\n Enter Number 1: ");
    scanf("%d",&x);
    printf("\n Enter Number 2: ");
    scanf("%d",&y);
    printf("\n Enter Number 3: ");
    scanf("%d",&z);
    printf("\n Average is: %d ",(x+y+z)/3);
    getch();
}
```

Output :



```
Enter Number 1: 23

Enter Number 2: 33

Enter Number 3: 43

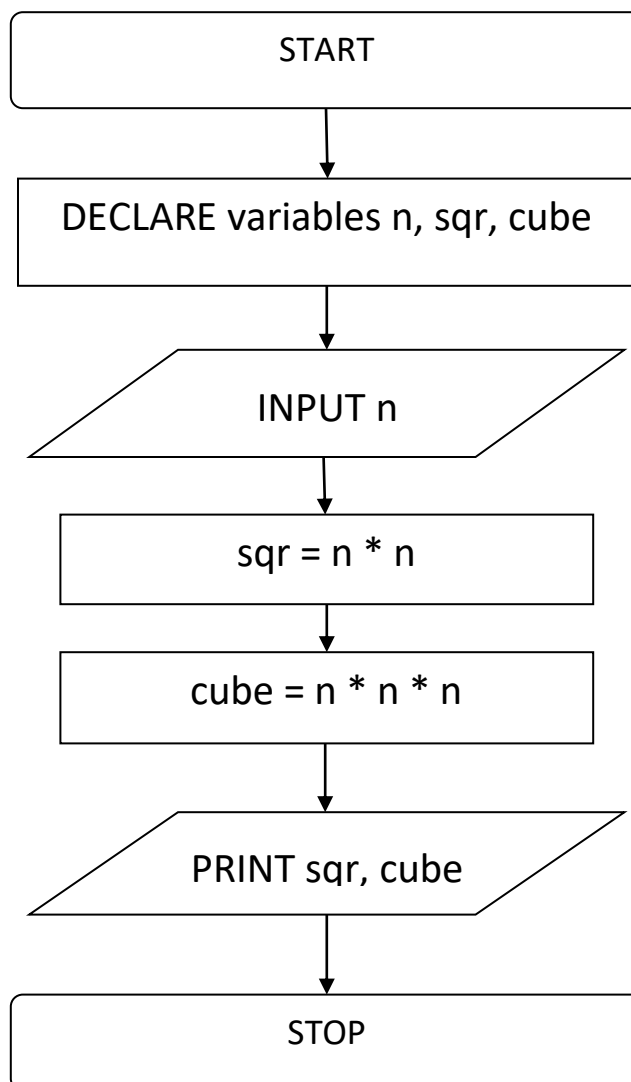
Average is: 33 _
```


(5) Calculate the Square and Cube of a number.

Algorithm :

Step 1 : START
Step 2 : DECLARE variables n , sqr , $cube$
Step 3 : INPUT n
Step 4 : $sqr = n * n$
Step 5 : $cube = n * n * n$
Step 6 : OUTPUT sqr , $cube$
Step 7 : STOP

Flowchart :

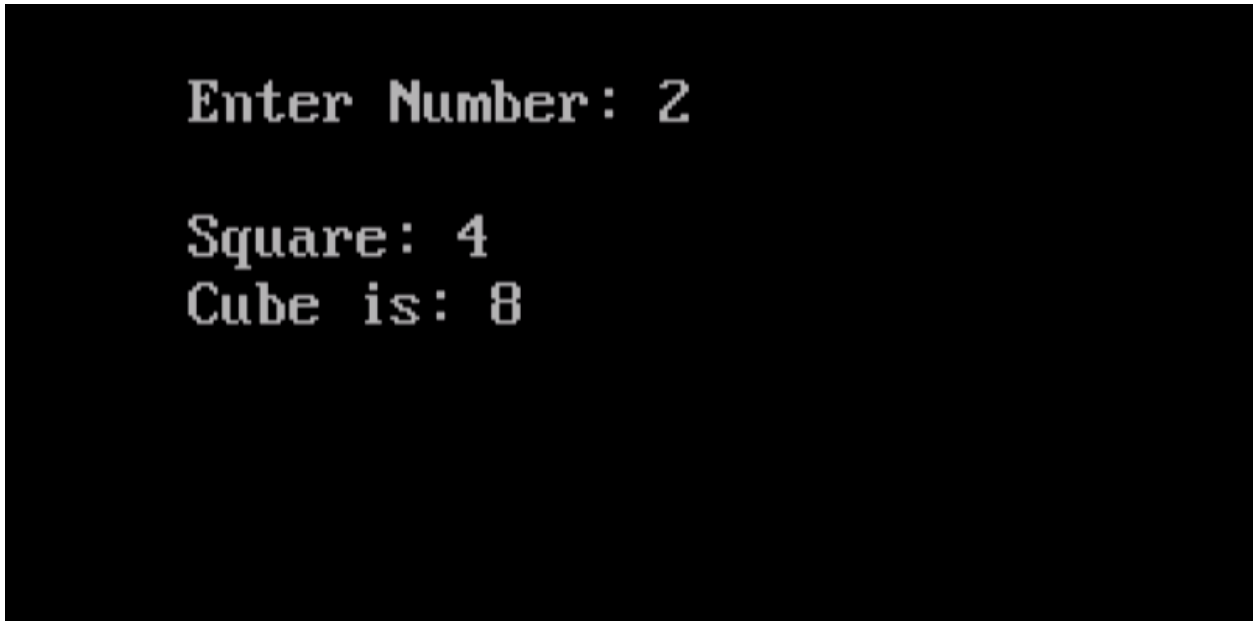


Code :

```
#include<stdio.h>
#include<conio.h>

void main()
{
    int n;
    clrscr();
    printf("\n Enter Number: ");
    scanf("%d",&n);
    printf("\n Enter Square: %d",n*n);
    printf("\n Cube is: %d",n*n*n);
    getch();
}
```

Output :



Enter Number: 2

Square: 4

Cube is: 8

(6) Interchange the value of two numbers.

Algorithm :

Step 1 : START

Step 2 : DECLARE variables temp, x, y

Step 3 : INPUT x, y

Step 4 : temp = x

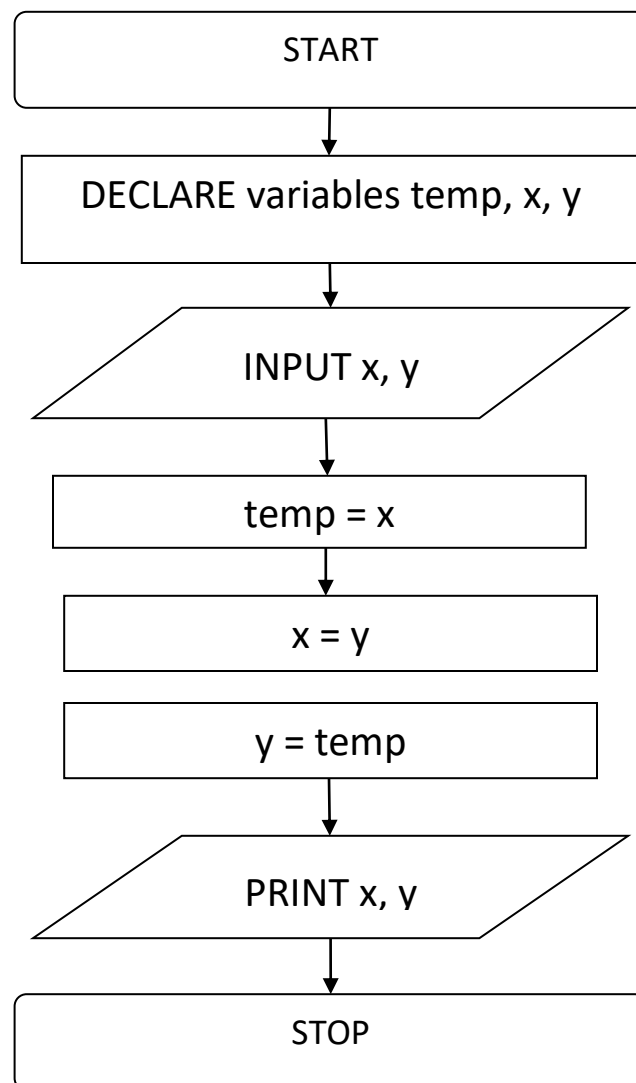
Step 5 : x = y

Step 6 : y = temp

Step 7 : PRINT x, y

Step 8 : STOP

Flowchart :

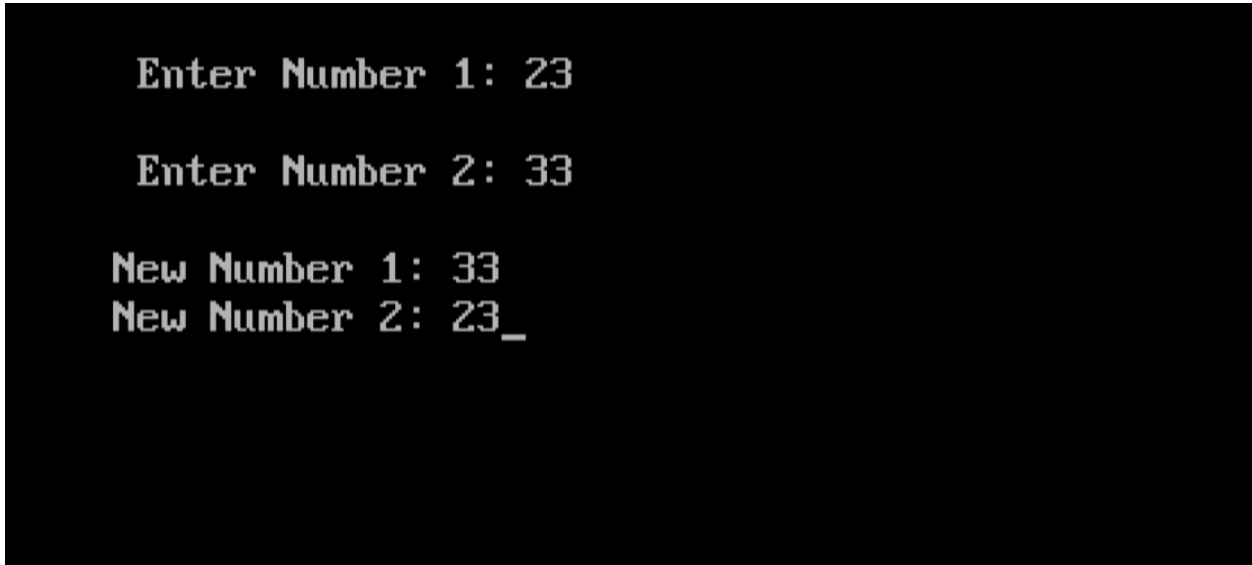


Code :

```
#include<stdio.h>
#include<conio.h>

void main()
{
    int temp,x,y;
    clrscr();
    printf("\n Enter Number 1: ");
    scanf("%d",&x);
    printf("\n Enter Number 2: ");
    scanf("%d",&y);
    temp=x;
    x=y;
    y=temp;
    printf("\nNew Number 1: %d",x);
    printf("\nNew Number 2: %d",y);
    getch();
}
```

Output :

A screenshot of a terminal window with a black background and white text. The output shows the program's execution: it prompts for 'Enter Number 1:' and receives '23', then prompts for 'Enter Number 2:' and receives '33'. It then displays 'New Number 1: 33' and 'New Number 2: 23_' where the underscore indicates the cursor is still on the line.

```
Enter Number 1: 23

Enter Number 2: 33

New Number 1: 33
New Number 2: 23_
```

(7) Find largest out of three numbers.

Algorithm :

Step 1 : START

Step 2 : DECLARE variables X, Y, Z

Step 3 : INPUT X, Y, Z

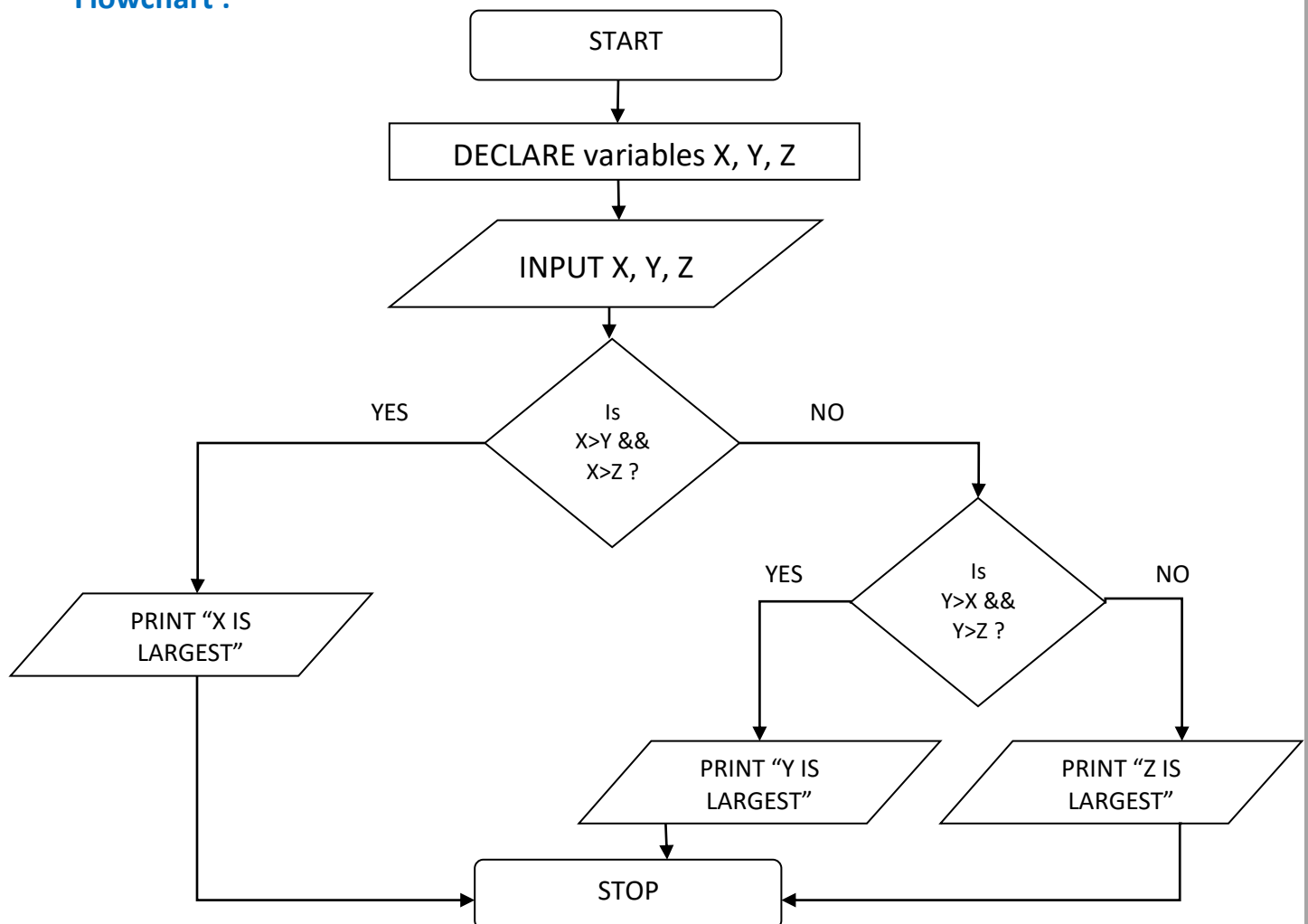
Step 4 : IF $X > Y$ && $X > Z$ THEN
PRINT "X is Largest"

Step 5 : ELSE IF $Y > Z$ && $Y > X$ THEN
PRINT "Y is Largest"

Step 6 : ELSE
PRINT "Z is Largest"

Step 7 : STOP

Flowchart :



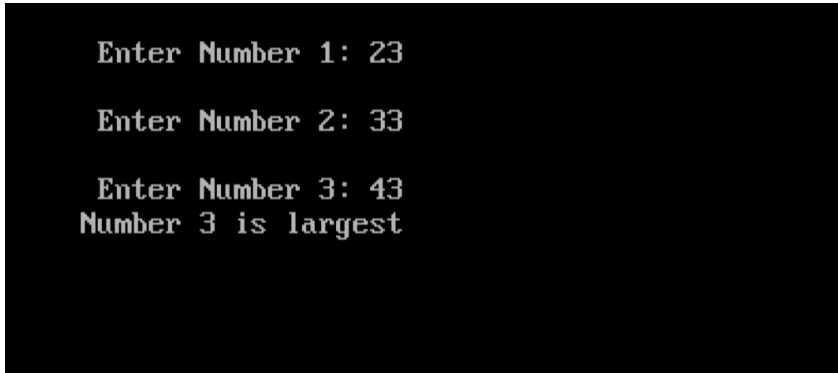
Code :

```
#include<stdio.h>
#include<conio.h>

void main(){
    int x,y,z;
    clrscr();
    printf("\n Enter Number 1: ");
    scanf("%d",&x);
    printf("\n Enter Number 2: ");
    scanf("%d",&y);
    printf("\n Enter Number 3: ");
    scanf("%d",&z);

    if(x>y && x>z){
        printf("Number 1 is largest");
    }
    else if(y>z && y>x){
        printf("Number 2 is largest");
    }
    else{
        printf("Number 3 is largest");
    }
    getch();
}
```

Output :

A screenshot of a terminal window with a black background and white text. It shows the output of the C program: 'Enter Number 1: 23', 'Enter Number 2: 33', 'Enter Number 3: 43', and 'Number 3 is largest'.

```
Enter Number 1: 23
Enter Number 2: 33
Enter Number 3: 43
Number 3 is largest
```

(9) Find out Number is Odd or Even.

Algorithm :

Step 1 : START

Step 2 : DECLARE variable $x = 0$

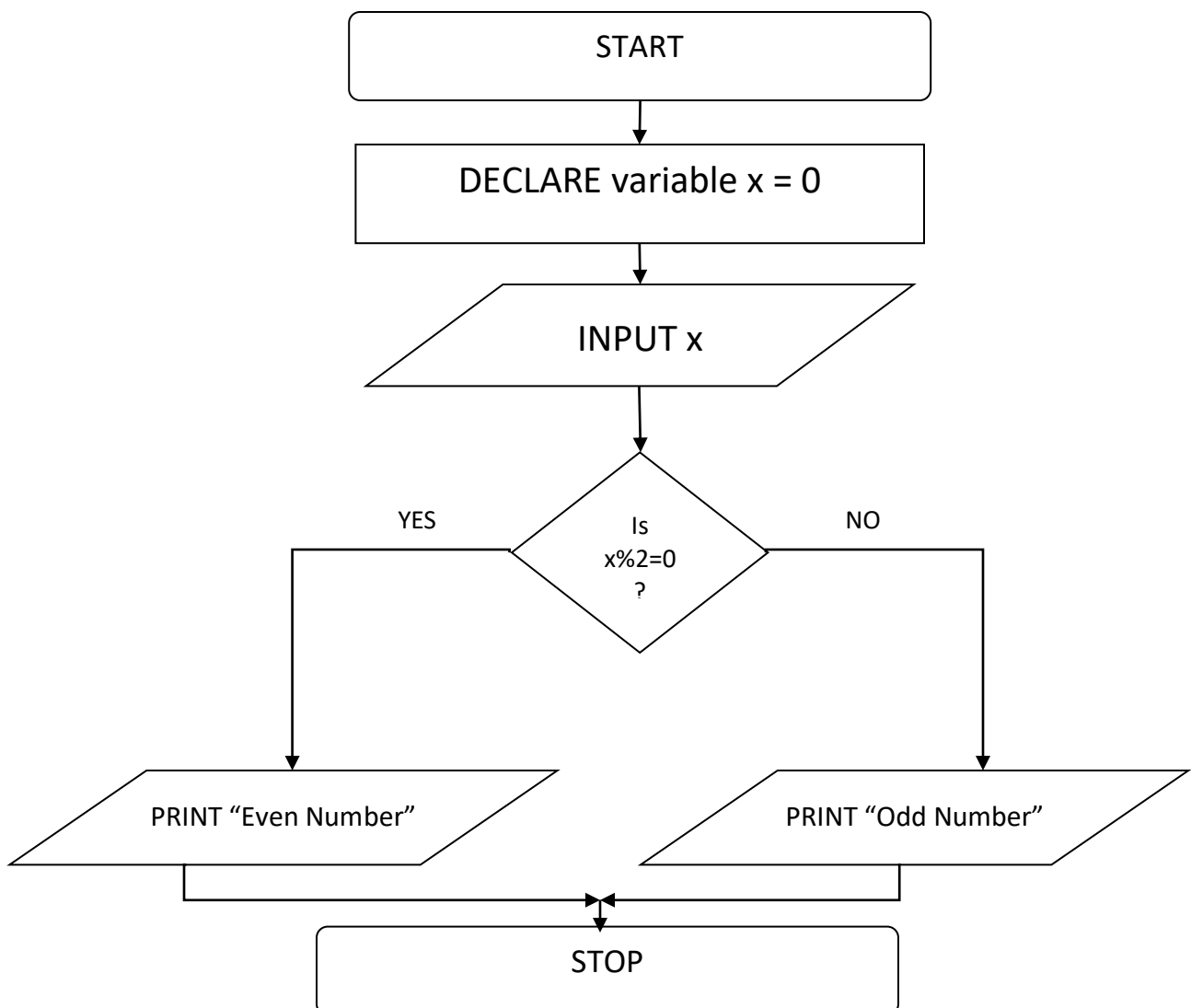
Step 3 : INPUT x

Step 4 : IF $x \% 2 = 0$ THEN
PRINT "Even Number"

Step 5 : ELSE
PRINT "Odd Number"

Step 6 : STOP

Flowchart :



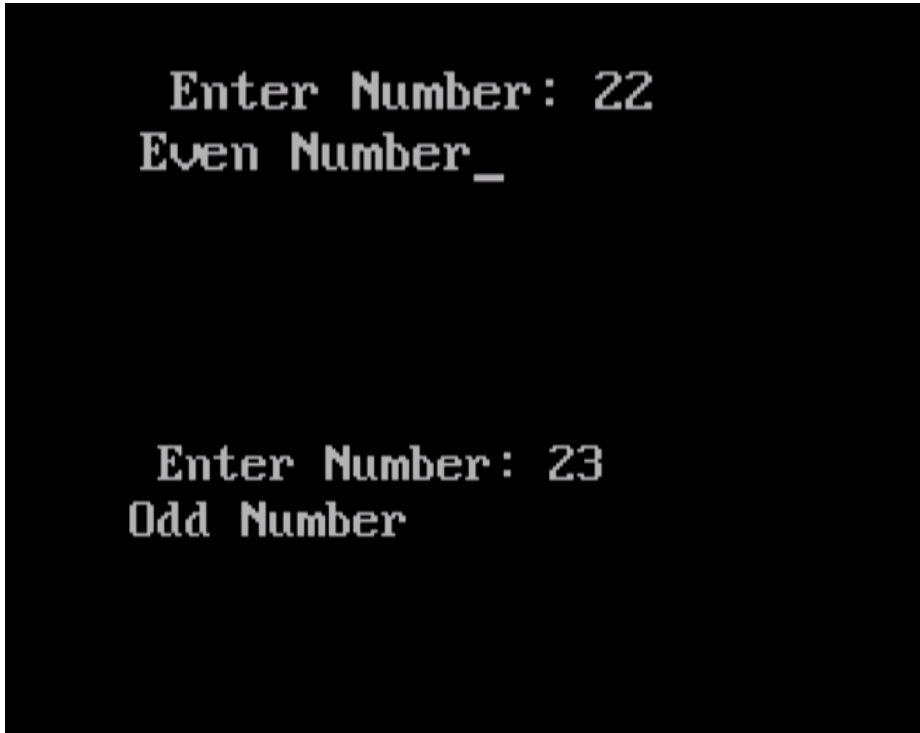
Code :

```
#include<stdio.h>
#include<conio.h>

void main()
{
    int n;
    clrscr();
    printf("\n Enter Number: ");
    scanf("%d",&n);

    if(n%2==0){
        printf("Even Number");
    }
    else
        printf("Odd Number");
    getch();
}
```

Output :



Enter Number: 22
Even Number_

Enter Number: 23
Odd Number

(10) Find out sum of digits of a given number.

Algorithm :

Step 1 : START

Step 2 : DECLARE variables n, x, y

Step 3 : INPUT n

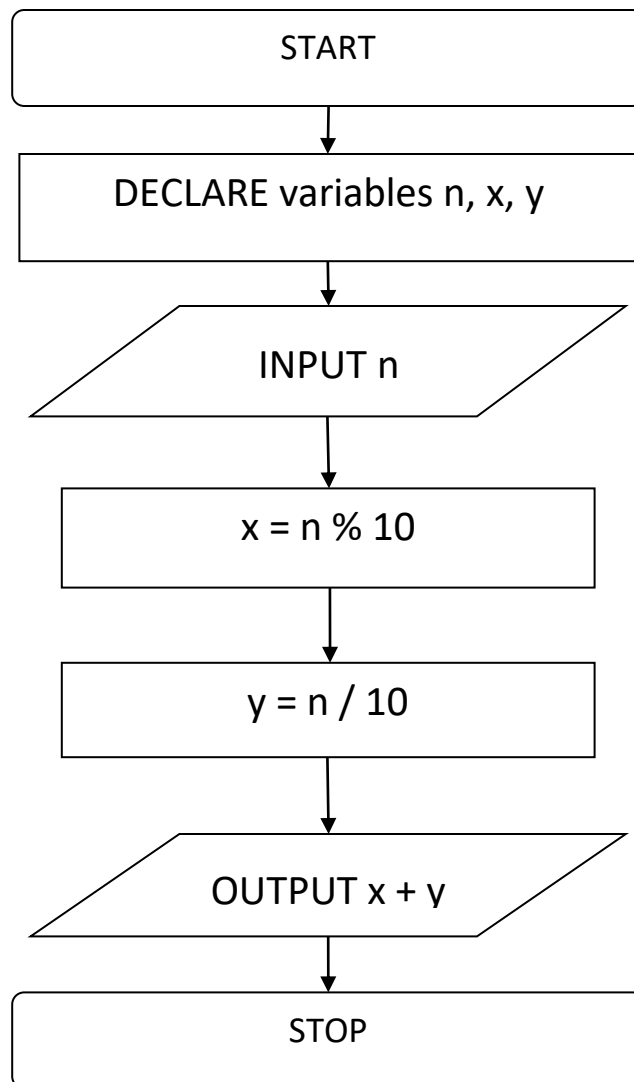
Step 4 : $x = n \% 10$

Step 5 : $y = n / 10$

Step 6 : OUTPUT $x + y$

Step 7 : STOP

Flowchart :

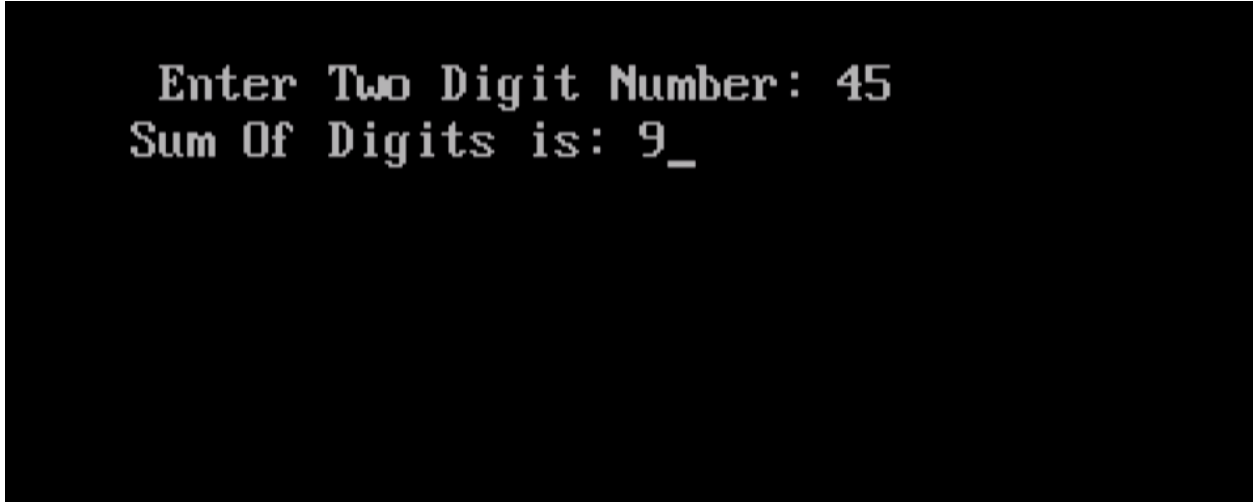


Code :

```
#include<stdio.h>
#include<conio.h>

void main()
{
    int n,x,y;
    clrscr();
    printf("\n Enter Two Digit Number: ");
    scanf("%d",&n);
    x=n%10;
    y=n/10;
    printf("Sum Of Digits is: %d",x+y);
    getch();
}
```

Output :



```
Enter Two Digit Number: 45
Sum Of Digits is: 9_
```